



APPLICATIONS OF **ADVANCED** ✨ ✨ ✨ **NANOMATERIALS**

Tufan Ghosh | Roopas Kiran Sirugudu | Sudagar Jothi



Applications of Advanced Nanomaterials



— *Editor* —

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Chapter 1

The Influence of Organic Dye on the Electrical Properties of n-Ge Based Organic-on-Inorganic Semiconductor Heterostructures

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INTRODUCTION

As the scaling of Si-based technology faces critical issues, the germanium (Ge) is considered as an alternative material in next-generation complementary metal oxide semiconductor (CMOS) devices to push forward the limitations of Si-based technology platforms due to its symmetric and high carrier mobility than Si. But strong Fermi-level pinning in the vicinity of the valence band maximum of Ge semiconductor raises the effective electron Schottky barrier height (SBH) of metal contacts on n-Ge [1–4]. Therefore, forming quality metal/Ge interface and ohmic contacts with n-Ge remains an obstacle in realizing high performance Ge-based CMOS devices. Several approaches have been proposed to overcome this obstacle, such as inserting interfacial layers (eg. Al_2O_3 , TiO_2 , SiO_2 , ZnO , MgO , Ge_3N_4 , TaN) between a metal and n-Ge [5-10]. Similarly, the introduction of an organic

interlayer to the metal/Ge interface is another approach to modulate the Schottky barrier height. For instance, we demonstrate the presence of a sulforhodamine-G (SRG) interlayer in the Au-Ge interface leads to the modification of the effective barrier height of Au/n-type Ge Schottky contact. Recently, Methylene Blue ($C_{16}H_{18}N_3ClS$), Orange-G ($C_{16}H_{10}N_2O_7S_2Na_2$), perylene monoamide ($C_{28}H_{24}N_2O_2$) have been considered as a very promising conducting organic dyes for various applications in electronic devices due to its superior properties such as high conductivity, structural stability, optical transparency, and processability [5,11-13]. Sulforhodamine-G is an ambipolar fluorescent dye with $C_{25}H_{25}N_2NaO_7S_2$ molecular formula which found significant applications in organic electronics as an acceptor dye. The device structure for electrical characterisation and the chemical structure of sulforhodamine-G is shown in Fig. 1

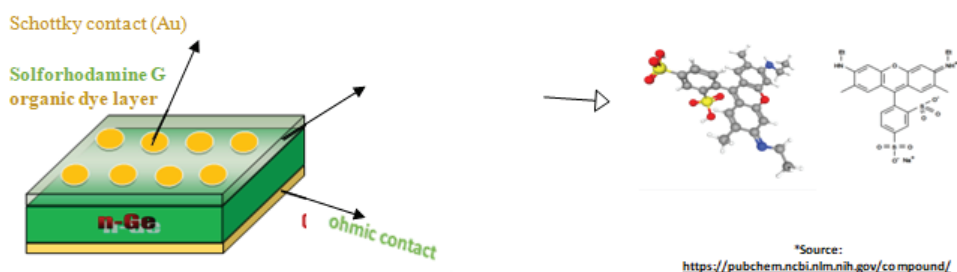


Fig. 1. Device structure for electrical characterization and Molecular structure (3D and 2D) of Sulforhodamine-G

Many researchers have studied about the influence of different interfacial layers on the performance of Au/n-Ge Schottky structures [14-18]. A low cost carbon paste thin film was coated on n-Ge and studied the electrical properties of the contacts. The use of carbon paste as an interlayer shows a highly stable and efficient contact and the device is found to operate even for 500h continuously without degrading its characteristics [14]. Kim et al [15] studies about the contact area dependent current transport mechanism of Au/n-Ge contacts. The Poole-Frenkel mechanism is more dominant for decreased contact area of the contacts. Yildirim fabricated Au/n-Ge contacts using Rubrene as interfacial layer [16]. The inclusion of Rubrene layer between Au and n-Ge shows improvement in device performance and enable high quality Schottky structures. Jyothi et al [17] studied about the electrical properties of Pt/pyronine-B/n-Ge heterostructure. The pyronine-B interlayer is coated on n-Ge using low cost spin coating method. The authors found improvement in device efficiency with pyronine-B interlayer and the current conduction mechanism at higher bias voltages are dominated by Poole-Frenkel emission due to the formation of defects or traps below the conduction band of n-Ge due to organic interlayer. A study on Schottky barrier properties of Au/n-Ge contacts with monolayer grapheme demonstrated the enhancement of contact parameters such as low ideality factor, low series resistance and higher barrier height than compared to Au/n-Ge conventional contacts [18]. The interfacial graphene layer not only passivates the semiconductor surface but also reduces the interface state density. Still the new materials show

the pathway to understand the enhancement of electrical properties and current conduction mechanisms of various heterostructures using different interfacial layers.

In this work, we have fabricated Au/n-type Ge Schottky contact with a thin organic interlayer (SRG) using low-cost spin coating technique and demonstrated its electrical properties. Based on thermionic emission model, Schottky barrier parameters of Au/SRG/n-type Ge Schottky contact is evaluated using current-voltage (I-V) characteristics. The Cheung's and Norde methods are also utilised to evaluate the series resistance, barrier height and ideality factor to understand their consistency and validity. It was shown that SRG interlayer in-between metal and Ge substrate is effective in surface passivation of Ge and also influence the electrical properties of the conventional Au/n-Ge Schottky contacts.

MATERIALS AND METHODS

In this work, n-type Ge (100) (Sb-doped) wafers with a carrier concentration of $2 \times 10^{18} \text{ cm}^{-3}$ is used to fabricate Au/SRG/n-Ge heterojunction. The wafer was degreased in trichloroethylene, acetone and methanol. Then, it was etched by $\text{H}_2\text{O}:\text{HF}$ (100:1) solution to remove the native oxide from the surface of the wafer. Preceding each step, the wafer was rinsed in deionized water. Immediately after surface cleaning procedures, the sample was loaded into vacuum chamber to coat ohmic contact on the rough side of the sample with aluminum (Al) of 50 nm thickness. After ohmic contact formation, Sulforhodamine-G (SRG) film was coated with spin coater on the smooth side of the sample. Spin coater is operated at 4000 rpm with 60 sec duration to coat SRG on the smooth side of n-Ge surface and dried at 150°C for 30 min. in ambient. Au thin film of 40 nm thickness was deposited using e-beam technique onto the top of the SRG thin film using a shadow metal mask. For a comparison, the Au Schottky contact without a SRG interlayer was fabricated on the n-type Ge wafer using same process conditions. The evaporation process was carried out at 5×10^{-6} Torr. The electrical characteristics of the Au/SRG/n-Ge and Au/n-Ge Schottky contacts were measured at room temperature using a precision semiconductor parameter analyzer.

RESULTS AND DISCUSSION

Fig. 2 shows the I-V characteristics of Au/n-type Ge Schottky contact with and without SRG interlayer measured at room temperature. It is clear that both devices exhibited good rectifying properties. Based on thermionic emission model, the forward bias I-V characteristics of Schottky contact with the series resistance can be expressed as:

$$I_o = A A^{**} T^2 \exp\left(-\frac{q \phi_b}{KT}\right) \quad (1)$$

where q is the electron charge, V is the applied voltage, k is the Boltzmann constant, T is the absolute temperature, R_s is the series resistance, n is the ideality factor, I_o is the saturation current and can be obtained from the intercept of the plot of $\ln(I)$ versus V at $V = 0$ and the ideality factor ' n ' can be determined from the slope of the linear region of the forward bias $\ln(I)$ vs. (V) graph expressed as:

$$I_o = AA^{**}T^2 \exp\left(\frac{q\Phi_b}{KT}\right) \quad (2)$$

where A is the area of the diode, A^{**} is the Richardson constant ($140 \text{ A cm}^{-2} \text{ K}^{-2}$ for n-type Ge) and Φ_b is the barrier height. Once the saturation current I_o is determined, the barrier height Φ_b can be expressed as:

$$\Phi_b = \frac{kT}{q} \ln\left(\frac{AA^{**}T^2}{I_o}\right) \quad (3)$$

The barrier height and ideality factor were found to be 0.63 eV and 1.51 for the Au/n-type Ge Schottky contact, and 0.73 eV and 1.30 for the Au/SRG/n-Ge Schottky contact, respectively. Both devices exhibited an ideality factor deviating from the value of unity. This could be associated with the presence of a wide distribution of low-Schottky barrier patches caused by a laterally inhomogeneous barrier [19-20]. It is also evidenced that the barrier height of Au/SRG/n-Ge Schottky contact was higher than that of Au/n-type Ge Schottky contact. This indicates that the SRG interlayer effectively modulated the barrier height by influencing the interfacial potential barrier of the Au/Ge Schottky junction. The image-force effect, recombination-generation and tunnelling are all possible mechanisms that could lead to an ideality factor greater than unity. The Cheung's method [21] was employed to calculate the barrier height, ideality factor, and series resistance of Au/n-type Ge Schottky contact with and without SRG interlayer.

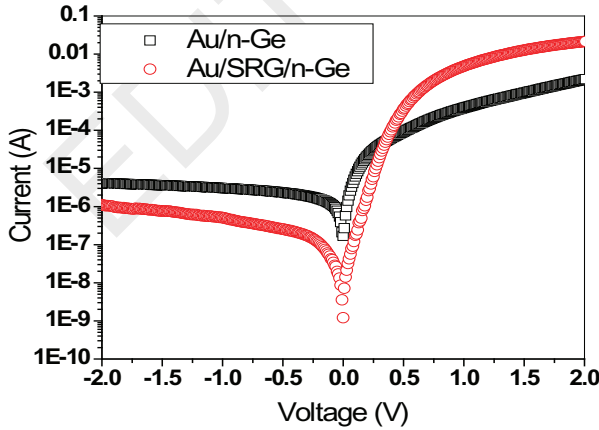


Fig. 2. I-V characteristics of Au/n-type Ge and Au/SRG/n-type Ge Schottky contact measured at room temperature.

Figure 3 shows a plot of $dV/d(\ln I)$ versus I and $H(I)$ versus I for the Au/n-type Ge and Au/SRG/n-Ge Schottky contact. The series resistance and ideality factor were extracted from the slope and y-axis intercept of the plot respectively and obtained as 3027 and 1.97 for the Au/n-type Ge Schottky contact and 69 and 2.59 for the Au/SRG/n-type Ge Schottky contact. Using the value of ideality factor determined from the plot of $dV/d(\ln I)$ versus I , the $H(I)$ versus I plot gives a straight

line with the y-axis intercept yielding the barrier height. The slope of $H(I)$ versus I plot also provides a second method to determine the series resistance. The series resistance and barrier height extracted from the plot of $H(I)$ versus I plot were found to be 2974 Ω and 0.61 eV for the Au/n-type Ge Schottky contact, and 53 Ω and 0.69 eV for the Au/SRG/n-type Ge Schottky contact, respectively. For both devices, the values of series resistance calculated from the plots of $dV/d(\ln I)$ versus I were almost identical to those from the plots of $H(I)$ versus I .

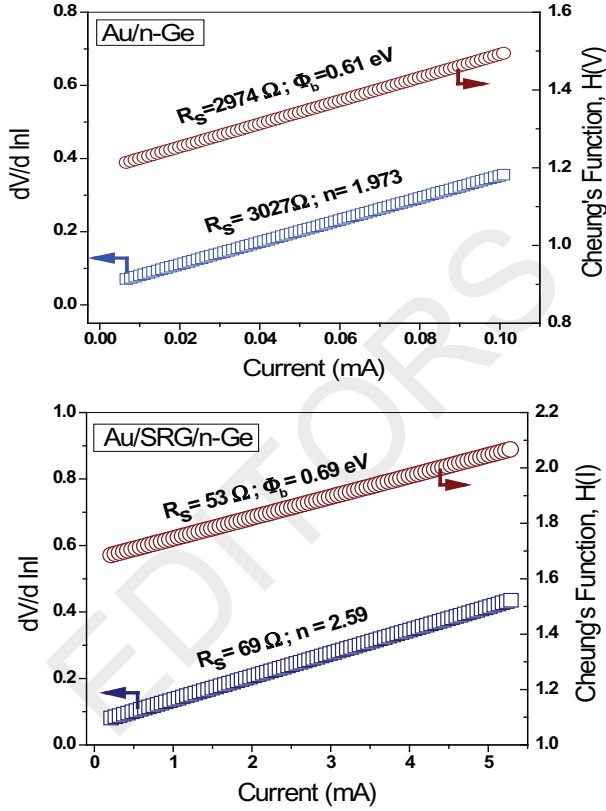


Fig. 3. Plot of $dV/d(\ln I)$ versus I and $H(I)$ versus I of Au/n-Ge and Au/SRG/n-Ge Schottky contact.

Norde proposed an empirical function to calculate the barrier height and series resistance for a Schottky contact. The Norde function is defined as [22]:

$$F(V) = \frac{V}{\gamma} - \frac{kT}{q} \ln \left(\frac{I(V)}{A A^{**} T^2} \right) \quad (4)$$

where γ is an integer (dimensionless) greater than the ideality factor and $I(V)$ is the current obtained from the forward bias I - V characteristics of the contacts. The basic technique of Norde method is to plot the proposed function with respect to the voltage applied across the Schottky contact, and to obtain its minimum. Once

the minimum of the $F(V)$ versus V plot is determined, the value of barrier height can be obtained using the following equation:

$$\Phi_b = F(V_{\min}) + V_{\min} - \frac{kT}{q} \quad (5)$$

where $F(V_{\min})$, $V_{\min}$, and $I_{\min}$ are the minimum value of $F(V)$, the corresponding voltage and current, respectively. From the Norde function, the value of series resistance can be determined using the following equation:

$$R_s = \frac{kT}{q} \frac{(\gamma - n)}{I_{\min}} \quad (6)$$

Figure 4 shows the plot of $F(V)$ versus V of Au/n-type Ge Schottky contact with and without SRG interlayer. The barrier height and series resistance, extracted from the Norde plot, were found to be 0.63eV and 2458 Ω for the Au/n-type Ge Schottky contact, and 0.75eV and 73 Ω for Au/SRG/n-Ge Schottky contact, respectively. As shown in Table 1, a good consistency is observed between the values of barrier height, ideality factor and series resistance of Au/n-type Ge Schottky contact obtained using different approaches.

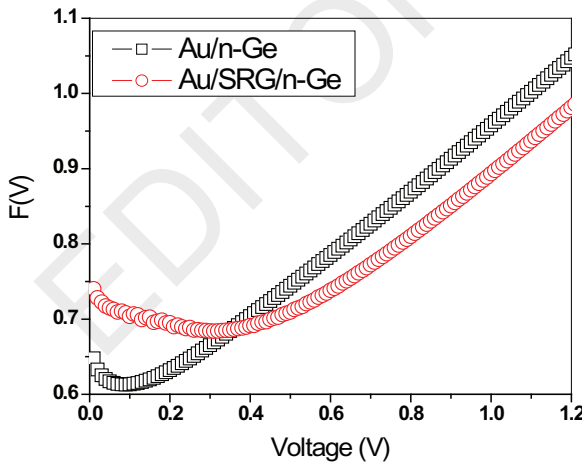


Fig. 4. Plot of $F(V)$ versus V obtained from I-V characteristics of Au/n-type Ge and Au/SRG/n-type Ge Schottky contact.

Cheung's method and the Norde method, implying their validity for the evaluation of Schottky barrier parameters for the contacts with and without SRG interlayer. However, differences in the barrier height values obtained from these different techniques for the present devices could be attributed to the extraction parameters from different regions of the forward bias I-V plot [23-24]. For instance, $\ln(I) - V$ method and Cheung method are only applied to extract barrier height in the linear (low voltage) and non-linear (high voltage) regions of the forward bias I-V characteristics, respectively. Besides, the barrier height was calculated from the full forward bias I-V characteristics of the Schottky contact in Norde method.

The current conduction mechanisms of Au/n-Ge contacts with and without SRG interlayer was analysed using double logarithmic scale forward bias I-V characteristics. The double logarithmic scale I-V data is fitted linearly ($I \propto V^m$) to identify the dominant current conduction at different bias regions for different magnitudes of slope factors ('m'). The observed dominant current conduction mechanism in Au/n-Ge contacts is due to ohmic conduction with slope factor $m=1.59$. on the other hand, three linear regions are observed in Au/SRG/n-Ge contacts. At low bias regions, the current-voltage follows a linear relation with slope ($m=1.282$) might be due to ohmic conduction but at moderately high voltage regions the slope is raised ($m=5.546$). The dominant current conduction mechanism observed is trap assisted space charge limited current in Au/SRG/n-Ge contacts at high bias regions. This might be due to the SRG interlayer introduces traps or defects at the interface below the conduction band of n-Ge semiconductor. Hence the current in these contacts are significantly controlled by traps or defects at the active layer of the Au/SRG/n-Ge contacts [16, 17]. Finally at high voltages, the evaluated slope factor found to be $m=2.086$, which might be due to trap-free space charge limited current conduction mechanism.

Table 1. Schottky barrier parameters extracted by using various methods of Au/n-type Ge and Au/SRG/n-type Ge Schottky contact.

Method	Sample	Ideality factor	Barrier height	Series resistance
I-V graph	Au/n-Ge	1.51	0.63 eV	—
	Au/SRG/n-Ge	1.30	0.73 eV	—
Cheung's Method ($dV/d\ln I$) vs. I	Au/n-Ge	1.97	—	3027
	Au/SRG/n-Ge	2.59	—	69
Cheung's Method ($H(I)$ Vs. I)	Au/n-Ge	—	0.61 eV	2974
	Au/SRG/n-Ge	—	0.69 eV	53
Norde Method	Au/n-Ge	—	0.63 eV	2458
	Au/SRG/n-Ge	—	0.75 eV	73

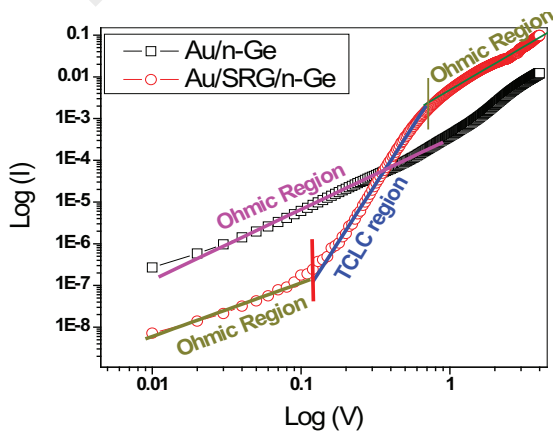


Fig. 5. Current conduction mechanism of Au/n-Ge and Au/SRG/n-Ge contacts in the Forward I-V characteristic.

CONCLUSION

In this work, we have investigated the electrical characteristics of the Au/SRG/n-Ge Schottky structures fabricated by spin coating of the organic dye material on n-Ge substrate. It has been seen that the SRG thin film on n-Ge substrate showed a good rectifying behaviour. The barrier height and the ideality factor of the device were calculated from the semi log I–V based on thermionic emission model. The Schottky barrier parameters such as barrier height, series resistance and ideality factor of the Au/SRG/n-Ge Schottky diodes compared with those of conventional Au/n-Ge contacts. The barrier height value of 0.73 eV obtained for the Au/SRG/n-Ge contact found significantly larger than the barrier height value of the conventional Au/n-Ge contact. Thus, the modification of the interfacial potential barrier for metal/Ge diodes has been achieved by using an SRG organic interlayer. This has been attributed to the fact that the SRG interlayer increased the effective ϕ_b by influencing the space charge region of Ge. The insertion of organic dye (SRG) thin layer between Au/n-Ge contacts shows a trap assisted charge limited current at higher bias voltage due to the creation of defects or traps at the active layer of the contact below the conduction band.

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